

The effect of intensive diabetes treatment on the progression of diabetic retinopathy in insulin-dependent diabetes mellitus. The Diabetes Control ...

OBJECTIVE: To determine the magnitude of the decrease in the risk of retinopathy progression observed with intensive treatment and its relationship to baseline retinopathy severity and duration of follow-up. **DESIGN:** Randomized clinical trial, with 3 to 9 years of follow-up. **SETTING AND PATIENTS:** Between 1983 and 1989, 29 centers enrolled 1441 patients with insulin-dependent diabetes mellitus aged 13 to 39 years, including 726 patients with no retinopathy and a duration of diabetes of 1 to 5 years (primary prevention cohort) and 715 patients with very mild to moderate nonproliferative diabetic retinopathy and a duration of diabetes of 1 to 15 years (secondary intervention cohort). Ninety-five percent of all scheduled examinations were completed. **INTERVENTIONS:** Intensive treatment consisted of the administration of insulin at least three times a day by injection or pump, with doses adjusted based on self-blood glucose monitoring and with the goal of normoglycemia. Conventional treatment consisted of one or two daily insulin injections. **OUTCOME MEASURES:** Change between baseline and follow-up visits on the Early Treatment Diabetic Retinopathy Study retinopathy severity scale, assessed with masked gradings of stereoscopic color fundus photographs obtained every 6 months. **RESULTS:** Cumulative 8.5-year rates of retinopathy progression by three or more steps at two consecutive visits were 54.1% with conventional treatment and 11.5% with intensive treatment in the primary prevention cohort and 49.2% and 17.1% in the secondary intervention cohort. At the 6- and 12-month visits, a small adverse effect of intensive treatment was noted ("early worsening"), followed by a beneficial effect that increased in magnitude with time. Beyond 3.5 years of follow-up, the risk of progression was five or more times lower with intensive treatment than with conventional treatment. Once progression occurred, subsequent recovery was at least two times more likely with intensive treatment than with conventional treatment. Treatment effects were similar in all baseline retinopathy severity subgroups. **CONCLUSIONS:** The results of

the Diabetes Control and Complications Trial strongly support the recommendation that most patients with insulin-dependent diabetes mellitus use intensive treatment, aiming for levels of glycemia as close to the nondiabetic range as is safely possible.